

Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 100506

Subject: Power Electronics

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):

[2 x 7 = 14]

- (a) Which device combines MOSFET input characteristics with BJT output characteristics?

(i) SCR	(iii) Diode
☒ (ii) IGBT	(iv) TRIAC
- (b) Turn-on time for an SCR is 10 μ sec. If an inserted in the anode circuit, then the turn-on time will be

(i) 10 μ sec	(iii) less than 10 μ sec
☒ (ii) more than 10 μ sec	(iv) about 10 μ sec
- (c) In a 3-phase semi-converter, for firing angle less than or equal to 60° , each thyristor and diode conduct, respectively, for

(i) $60^\circ, 60^\circ$	(iii) $90^\circ, 30^\circ$
☒ (ii) $120^\circ, 120^\circ$	(iv) $180^\circ, 180^\circ$
- (d) In a single-phase full converter, if output voltage has peak and average values of 325 V and 133 V respectively, then the firing angle is

(i) 40°	(iii) 140°
☒ (ii) 50°	(iv) 130°
- (e) A boost chopper is fed from a 220 V dc source to deliver a load voltage of 660 V. If the non-conduction time of the thyristor is 100 μ s, the required pulse width would be

(i) 100 μ s	☒ (iii) 200 μ s
(ii) 220 μ s	(iv) 660 μ s
- (f) A chopper can be used on

(i) Pulse-width modulation Only	(iii) Frequency modulation only
(ii) Amplitude modulation only	☒ (iv) Both PWM and FM
- (g) A single-phase full bridge VSI has inductor L as the load. For a constant source voltage, the current through the inductor is

(i) Square wave	(iii) Triangular wave
☒ (ii) Step function	☒ (iv) Pulsed wave
- (h) A single-phase full bridge inverter can operate in load-commutation mode in case load consists of

(i) RL	☒ (iii) RLC underdamped
(ii) RLC overdamped	(iv) RLC critically damped
- (i) The power delivered to a star-connected load of $R \Omega$ per phase, from a 3-phase bridge inverter fed from fixed dc source, is 10 kW for 180° mode. From 120° mode, the power delivered to load would be

(i) 10 kW	(iii) 5 kW
☒ (ii) 6.667 kW	(iv) 7.5 kW

- (j) The output voltage waveform of a 3-phase square-wave inverter contains
 (i) only even harmonics
 (ii) only odd harmonics
 (iii) both odd and even harmonics
 (iv) only triplan harmonics
- Q.2** (a) Explain the construction, working principle, and V-I characteristics of a SCR. [7]
 (b) Explain voltage commutation with suitable circuit diagram and waveforms. [7]
- Q.3** (a) Describe the operation and characteristics of a MOSFET used in power electronics applications. [7]
 (b) Explain RC firing circuit for SCR with appropriate waveforms. [7]
- Q.4** (a) Describe the working of a single-phase full converter in the inverter mode with RLE load. Illustrate your answer with waveforms for source voltage, E , load voltage and current, source current, current through and voltage across the SCR. Assume continuous conduction. [7]
 (b) A resistive load of $10\ \Omega$ is connected through a half-wave SCR circuit to 220 V, 50 Hz, single-phase source. Calculate the power delivered to load for a firing angle of 60° . Find also the value of input power factor. [7]
- Q.5** (a) A single-phase semi-converter feeds power to RLE load. For discontinuous load current, draw the source voltage, output voltage, load current, source current and freewheeling diode current waveforms as a function of time when
 (i) Extinction angle $\beta > \pi$
 (ii) Extinction angle $\beta < \pi$ with $V_m \sin \beta < E$ [7]
 (b) A 3-phase full converter is used for charging a battery with an emf of 110 V and an internal resistance of $0.2\ \Omega$. For a constant charging current of 10 A, compute the firing angle delay for ac line voltage of 220 V. Find also the supply power factor. [7]
- Q.6** (a) Explain the operation of a buck converter with circuit diagram and steady-state waveforms. Also derive the relationship between duty ratio and output voltage in a buck converter. [7]
 (b) A chopper fed from a 220 V dc source, is working at a frequency of 50 Hz and is connected to an R-L load of $R=5\ \Omega$ and $L=40\text{ mH}$. Determine the value of duty cycle at which the minimum load current will be (i) 5A (ii) 10 A (iii) 20 A and (iv) 30 A. [7]
- Q.7** (a) Describe the principal of boost converter. Drive an expression for the average output voltage in terms of input voltage and duty cycle. State the assumption made. [7]
 (b) Explain the operation of a single-phase voltage source inverter under square wave operation. [7]
- Q.8** (a) Explain the operation of a three-phase voltage source inverter with switch states and output waveforms for 120° mode of conduction. [7]
 (b) A single-phase full-bridge inverter delivers power to RLC load with $R=3\ \Omega$, $X_L=12\ \Omega$. The bridge operates with a periodicity of 0.2 ms. Calculate the value of C so that load commutation is achieved for the thyristors. Turn-off time for thyristors is $12\ \mu\text{s}$. Factor of safety is 2. Assume the load current to contain only the fundamental component. [7]
- Q.9** Write short note on **any two** of the following: [7 x 2 = 14]
 (a) Gate driver circuit for MOSFET and IGBT
 (b) Three-phase half wave rectifier with RLE load
 (c) Time ratio control and frequency control in DC to DC Converter
 (d) Pulse width modulation techniques